

K-Means Clustering

Technical Notes

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1 Introduction

Many learning problems begin with a response variable. Given a set of predictors, the objective is to explain or predict an observed outcome. Clustering addresses a different problem. The data consist only of observations and their measured features, and the objective is to determine whether some structure can be found among them.

Suppose that a dataset contains N observations,

$$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N,$$

where each observation is represented by p features,

$$\mathbf{x}_i = \begin{bmatrix} x_{i1} & x_{i2} & \cdots & x_{ip} \end{bmatrix}^T \in \mathbb{R}^p.$$

No response variable is available to indicate how these observations should be organized. Nevertheless, some observations may be more similar to each other than to the rest of the data. If this structure exists, it can be useful to divide the observations into groups whose members

are relatively similar within each group and relatively different across groups. This is the basic clustering problem.

Clustering is therefore an **unsupervised learning** problem. Unlike classification, the groups are not known beforehand and there are no class labels from which they can be learned. The groups are inferred from the structure of the observations themselves.

This distinction is important. A cluster is not the same object as a class. A class has an externally defined meaning, whereas a cluster is the result of organizing observations according to a particular definition of similarity. Even when class labels happen to be available for evaluation, a clustering algorithm does not necessarily recover those classes.

Consider, for example, a dataset containing fraudulent and legitimate transactions. If the fraud labels are hidden and the transactions are clustered using only their measured features, there is no reason to assume that two resulting clusters will correspond to fraud and non-fraud. The dominant structure in the feature space may instead reflect transaction amounts, purchasing behaviour, geography, or some combination of the available variables.

This leads to a more fundamental question: what does it mean for two observations to be similar?

Different clustering methods answer this question in different ways. K-Means gives a particularly simple answer. Observations are represented in Euclidean space, each cluster is represented by a point called its **centroid**, and a good partition is one in which observations lie close to the centroid of the cluster to which they belong.

The simplicity of this idea is one of the reasons K-Means is widely used. The method reduces a potentially complicated dataset to K representative points and an assignment of each observation to one of them. At the same time, this simplicity introduces assumptions that must be understood. The meaning of distance depends on the scale of the variables, the method favours particular cluster geometries, the solution can depend on its initialization, and the number of clusters K must be specified in advance.

These properties make K-Means useful not only as a clustering algorithm but also as a clear setting in which to study several fundamental questions in unsupervised learning: how a clustering problem is formulated, how a partition can be represented by an objective function, how that objective can be optimized, and how a clustering result should be evaluated when no single correct partition is known.

2 The Clustering Problem

Let the dataset be represented by the matrix

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_N^T \end{bmatrix} \in \mathbb{R}^{N \times p}.$$

A clustering of these observations into K groups can be represented by the sets

$$C_1, C_2, \dots, C_K,$$

where each C_k contains the indices of the observations assigned to the k -th cluster.

For these sets to define a partition of the data, every observation must belong to one cluster,

$$\bigcup_{k=1}^K C_k = \{1, 2, \dots, N\}, \quad (1)$$

and no observation can simultaneously belong to two different clusters,

$$C_k \cap C_l = \emptyset, \quad k \neq l. \quad (2)$$

For standard K-Means, the clusters are also taken to be non-empty,

$$|C_k| > 0, \quad k = 1, \dots, K. \quad (3)$$

These conditions define what constitutes a valid partition, but they do not yet tell us which partition should be preferred.

This is the central difficulty of clustering. Without known labels, there is no direct target against which every possible partition can be compared. A clustering method must therefore introduce a criterion that formalizes what it means for a partition to be good.

A natural requirement is that observations assigned to the same cluster should be similar. This property is commonly referred to as **within-cluster compactness**. A second desirable property is that different clusters should be well separated. These two ideas are related but are not identical, and different clustering methods and validation measures formalize them in different ways.

K-Means is built primarily around the first idea. It seeks clusters whose observations are compact around representative points.

2.1 Representing a Cluster by a Prototype

Suppose that a cluster C_k is represented by a prototype $\mu_k \in \mathbb{R}^p$. The distance between an observation $\mathbf{x}_i$ and this prototype can be measured using the Euclidean norm,

$$\|\mathbf{x}_i - \mu_k\|_2 = \sqrt{\sum_{j=1}^p (x_{ij} - \mu_{kj})^2}. \quad (4)$$

For K-Means, the corresponding squared Euclidean distance is used,

$$\|\mathbf{x}_i - \mu_k\|_2^2 = \sum_{j=1}^p (x_{ij} - \mu_{kj})^2. \quad (5)$$

Squaring the distance does not change which of a fixed collection of prototypes is closest to an observation, since the square function is monotonically increasing for non-negative values. It does, however, give K-Means its particular optimization structure: the point that minimizes the sum of squared Euclidean distances within a cluster is its arithmetic mean. This result will be derived explicitly below.

The use of Euclidean distance is therefore not an incidental implementation choice. It is part of the definition of standard K-Means and is directly connected to the use of the mean as the cluster prototype.

To see the geometry intuitively, consider observations with two features. For a fixed collection of centroids, every point in the plane can be assigned to its nearest centroid. This divides the feature space into regions, one for each centroid. The boundaries between these regions are formed by points that are equally distant from neighbouring centroids.

This nearest-centroid geometry is one half of K-Means. The other half is determining where the centroids themselves should be located.

3 From Clustering to K-Means

Assume initially that the partition

$$C_1, C_2, \dots, C_K$$

is known. For each cluster C_k , we would like to choose a representative point μ_k that is as close as possible to the observations assigned to that cluster.

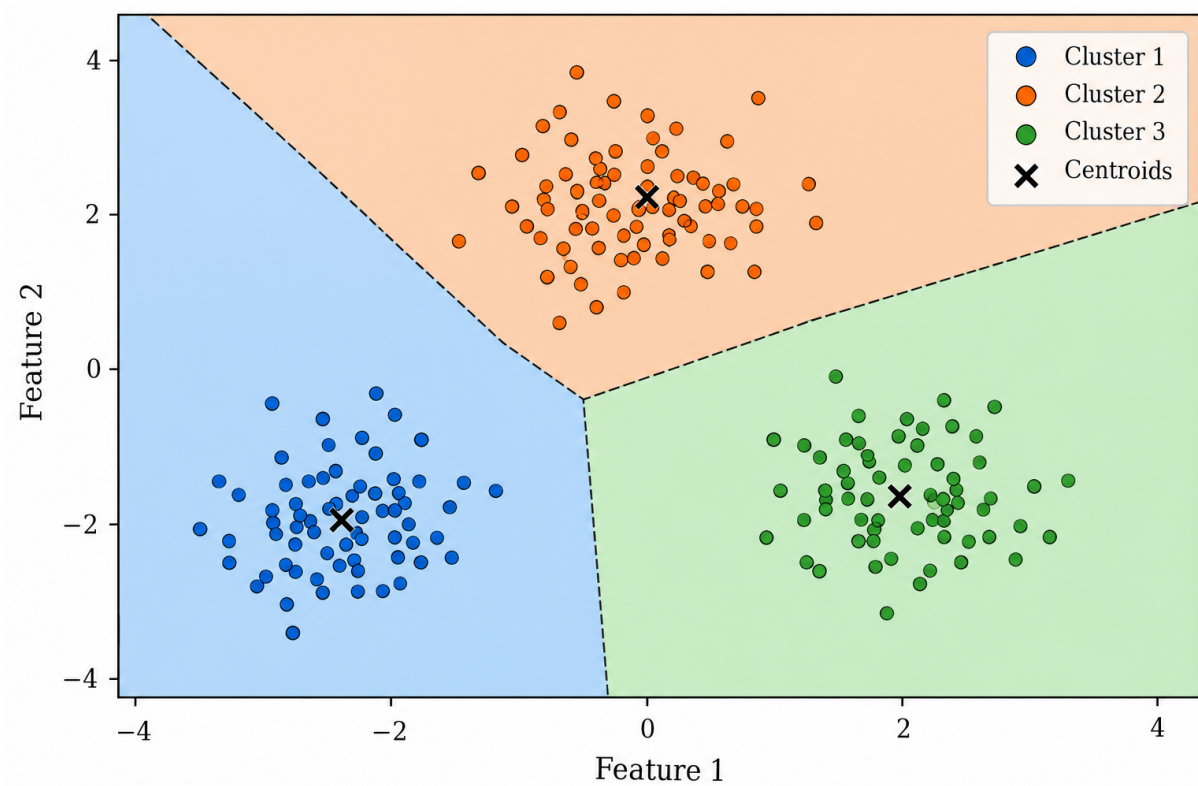


Figure 1: Nearest-centroid partition of a two-dimensional feature space. Each observation is assigned to its closest centroid, producing Voronoi regions whose boundaries consist of points equidistant from neighbouring centroids.

Using squared Euclidean distance, the total within-cluster variation around a candidate point $\mathbf{m}$ is

$$J_k(\mathbf{m}) = \sum_{i \in C_k} \|\mathbf{x}_i - \mathbf{m}\|_2^2. \quad (6)$$

The question is therefore

$$\arg \min_{\mathbf{m} \in \mathbb{R}^p} J_k(\mathbf{m}). \quad (7)$$

Let

$$\bar{\mathbf{x}}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i \quad (8)$$

be the arithmetic mean of the observations in C_k . For any candidate $\mathbf{m}$,

$$\mathbf{x}_i - \mathbf{m} = (\mathbf{x}_i - \bar{\mathbf{x}}_k) + (\bar{\mathbf{x}}_k - \mathbf{m}).$$

Taking the squared Euclidean norm gives

$$\begin{aligned} \|\mathbf{x}_i - \mathbf{m}\|_2^2 &= \|\mathbf{x}_i - \bar{\mathbf{x}}_k\|_2^2 + 2 (\mathbf{x}_i - \bar{\mathbf{x}}_k)^T (\bar{\mathbf{x}}_k - \mathbf{m}) \\ &\quad + \|\bar{\mathbf{x}}_k - \mathbf{m}\|_2^2. \end{aligned} \quad (9)$$

Summing over all observations in C_k ,

$$\begin{aligned} J_k(\mathbf{m}) &= \sum_{i \in C_k} \|\mathbf{x}_i - \bar{\mathbf{x}}_k\|_2^2 \\ &\quad + 2 \left[\sum_{i \in C_k} (\mathbf{x}_i - \bar{\mathbf{x}}_k) \right]^T (\bar{\mathbf{x}}_k - \mathbf{m}) \\ &\quad + |C_k| \|\bar{\mathbf{x}}_k - \mathbf{m}\|_2^2. \end{aligned}$$

The middle term vanishes because deviations from the mean sum to zero:

$$\begin{aligned}
\sum_{i \in C_k} (\mathbf{x}_i - \bar{\mathbf{x}}_k) &= \sum_{i \in C_k} \mathbf{x}_i - \sum_{i \in C_k} \bar{\mathbf{x}}_k \\
&= \sum_{i \in C_k} \mathbf{x}_i - |C_k| \bar{\mathbf{x}}_k \\
&= \sum_{i \in C_k} \mathbf{x}_i - |C_k| \left(\frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i \right) \\
&= \mathbf{0}.
\end{aligned} \tag{10}$$

Consequently,

$$J_k(\mathbf{m}) = \sum_{i \in C_k} \|\mathbf{x}_i - \bar{\mathbf{x}}_k\|_2^2 + |C_k| \|\bar{\mathbf{x}}_k - \mathbf{m}\|_2^2. \tag{11}$$

The first term does not depend on $\mathbf{m}$, while the second term is non-negative. Therefore,

$$J_k(\mathbf{m}) \geq \sum_{i \in C_k} \|\mathbf{x}_i - \bar{\mathbf{x}}_k\|_2^2,$$

with equality if and only if

$$\mathbf{m} = \bar{\mathbf{x}}_k.$$

Hence,

$$\boxed{\mu_k = \arg \min_{\mathbf{m} \in \mathbb{R}^p} \sum_{i \in C_k} \|\mathbf{x}_i - \mathbf{m}\|_2^2 = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i} \tag{12}$$

The arithmetic mean is therefore not chosen merely because it is a convenient summary statistic. It follows directly from the squared Euclidean objective.

For example, consider a one-dimensional cluster containing the observations

$$C_k = \{2, 4, 9\}.$$

Its centroid is

$$\mu_k = \frac{2 + 4 + 9}{3} = 5.$$

The within-cluster sum of squares at the centroid is

$$\begin{aligned} J_k(5) &= (2 - 5)^2 + (4 - 5)^2 + (9 - 5)^2 \\ &= 9 + 1 + 16 \\ &= 26. \end{aligned}$$

If another representative point is used, for example $m = 6$,

$$\begin{aligned} J_k(6) &= (2 - 6)^2 + (4 - 6)^2 + (9 - 6)^2 \\ &= 16 + 4 + 9 \\ &= 29. \end{aligned}$$

Equation Equation 11 explains the difference exactly:

$$J_k(6) = J_k(5) + 3(5 - 6)^2 = 26 + 3 = 29.$$

The result extends immediately to every cluster in the partition. Once the cluster assignments are fixed, the optimal representative of each cluster is its mean. Substituting these optimal representatives into the criterion gives the K-Means within-cluster sum of squares,

$$J(C_1, \dots, C_K) = \sum_{k=1}^K \sum_{i \in C_k} \|\mathbf{x}_i - \mu_k\|_2^2, \quad (13)$$

where

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i.$$

The K-Means problem can therefore be written as

$$\boxed{\text{minimize}_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{i \in C_k} \|\mathbf{x}_i - \mu_k\|_2^2} \quad (14)$$

subject to the partition conditions in Equation 1, Equation 2, and Equation 3.

This formulation makes the statistical meaning of K-Means precise. For a fixed value of K , the method searches for a partition whose observations are as compact as possible around their cluster means under squared Euclidean distance.

The definition is simple. Finding the globally optimal partition is not. Even a modest dataset can admit an enormous number of possible partitions. K-Means therefore relies on an iterative optimization procedure rather than enumerating every possible grouping. The standard procedure alternates between two operations: assigning observations to their nearest centroid and recomputing each centroid from the observations currently assigned to it. This is Lloyd’s algorithm and provides the computational link between the objective in Equation 14 and a practical clustering method.

4 References

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